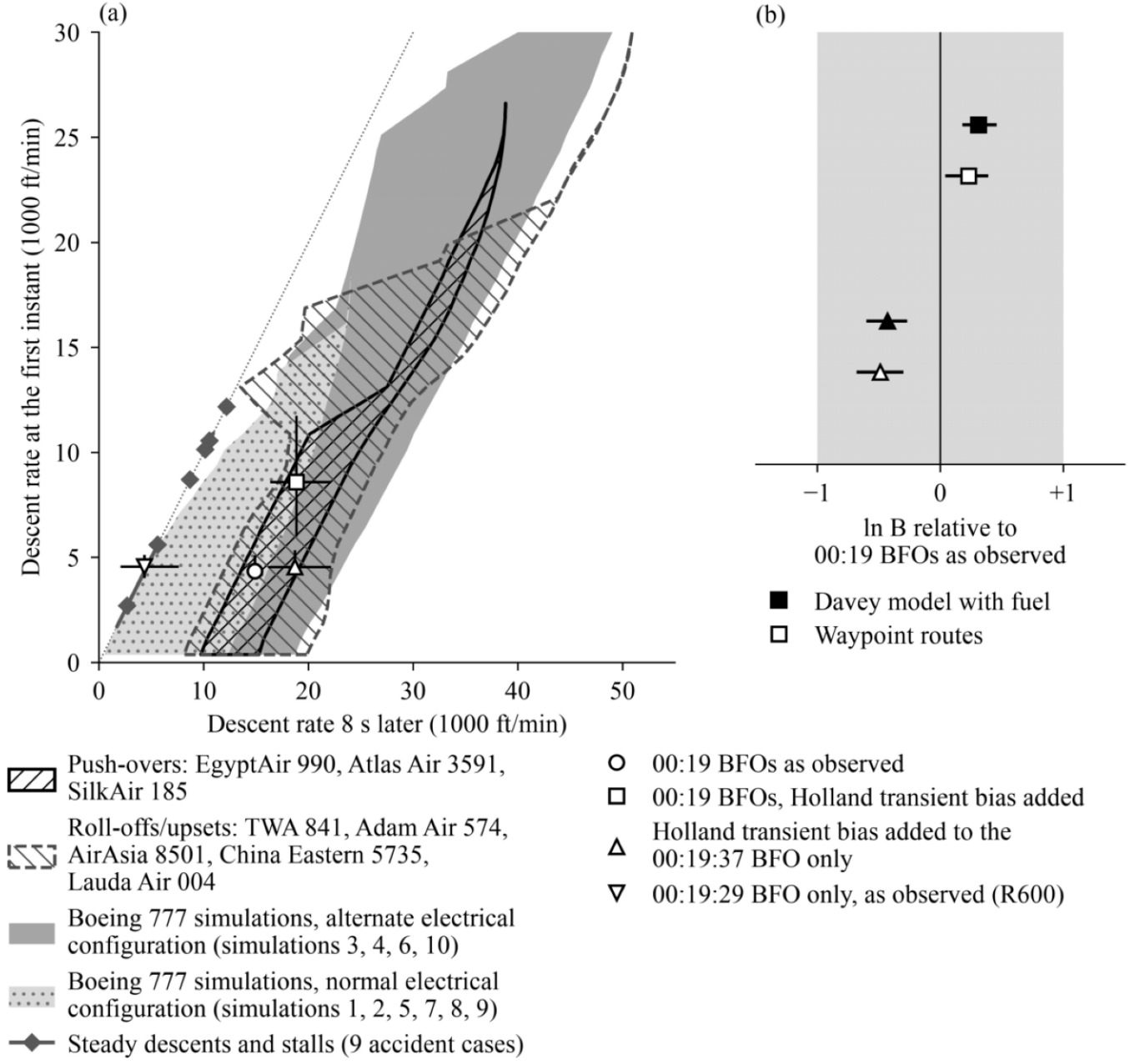
Candidate A1: descent-0019-A1-evidence-dots

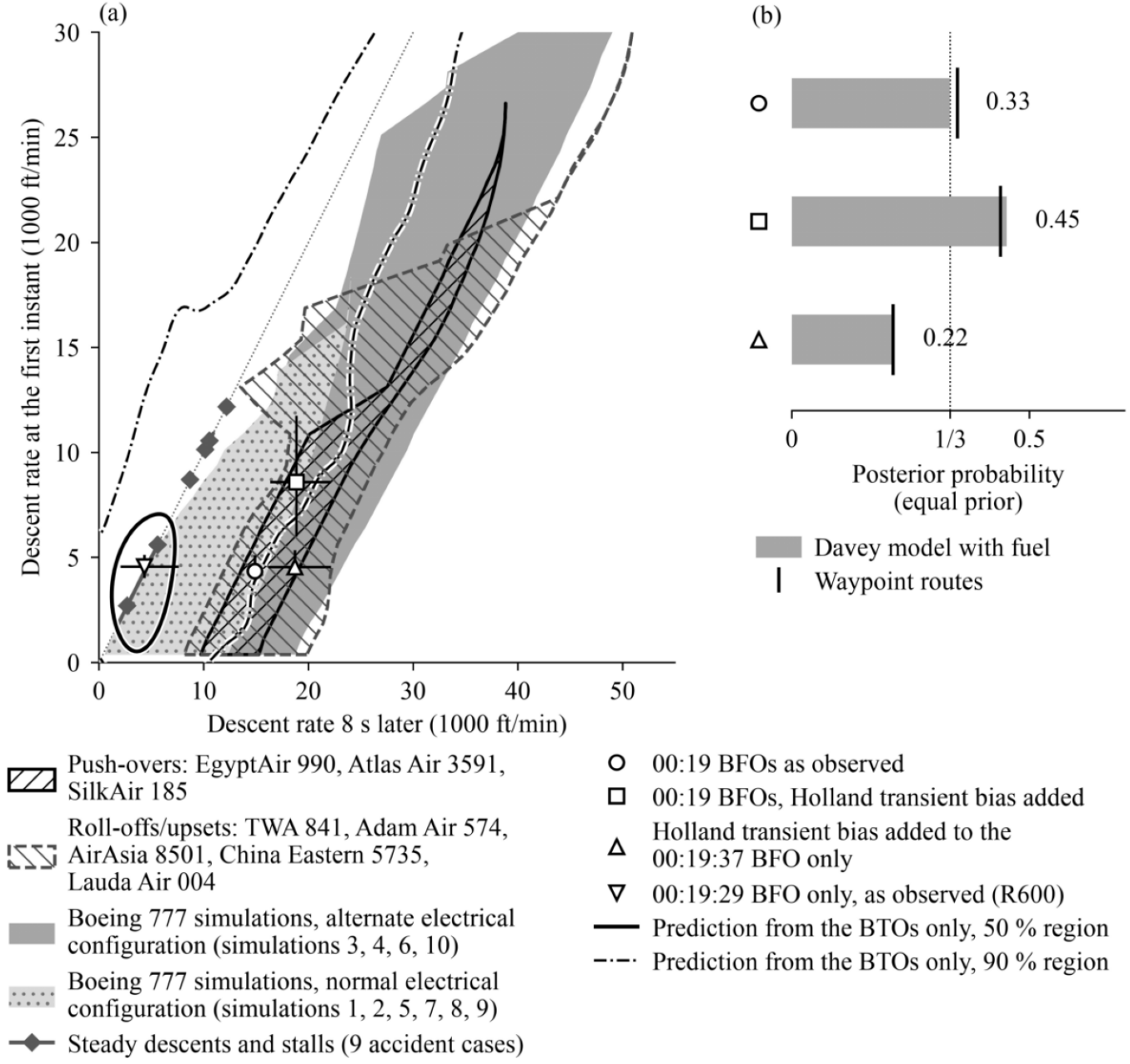
Log Bayes factor of the two bias treatments, both route models, with the |ln B| < 1 band. Strength: shows plainly that the evidence is weak. Weakness: covers only treatments with both BFOs; silent on the 18:25 log-on.



(a) Descent rate at a first instant and 8 s later during dive build-ups; each area covers one group of dives, and the dotted line is a steady descent. Symbols mark the 00:19:29 and 00:19:37 rates implied by each 00:19 BFO treatment; bars are 5–95 % posterior ranges (Davey model with fuel), centred on the published point for Holland transient bias. (b) Log Bayes factor against the 00:19 BFOs as observed, for the same data; lines are 90 % intervals over seeds. The grey band (|ln B| < 1) is evidence not worth more than a bare mention (Kass and Raftery, 1995). R600 is the 600 bit/s channel of the 00:19:29 burst.

Candidate A2: descent-0019-A2-prediction-contours

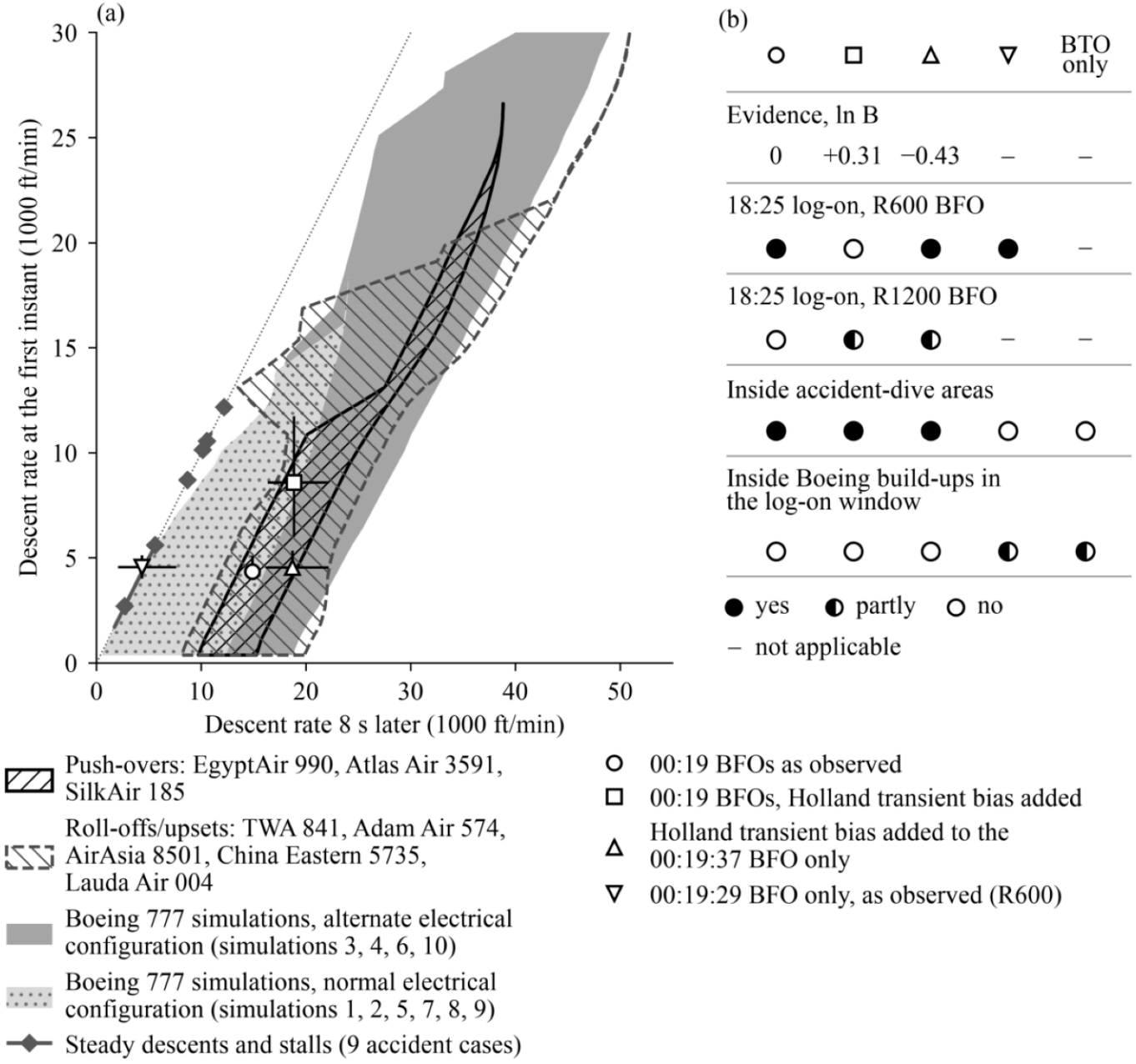
BTO-only prediction contours on the chart, and posterior probabilities. Strength: explains the small effective sample sizes. Weakness: two extra line styles on the main chart; the probabilities depend on the equal prior.



(a) Descent rate at a first instant and 8 s later during dive build-ups; each area covers one group of dives, and the dotted line is a steady descent. Symbols mark the 00:19:29 and 00:19:37 rates implied by each 00:19 BFO treatment; bars are 5–95 % posterior ranges (Davey model with fuel), centred on the published point for Holland transient bias. Contours enclose 50 % and 90 % of the rates predicted from the 00:19 BTOs alone for descending paths (27 % of the predicted weight). (b) Posterior probability, with an equal prior (dotted line), of the three treatments that use both BFOs. R600 is the 600 bit/s channel of the 00:19:29 burst.

Candidate A3: descent-0019-A3-scorecard

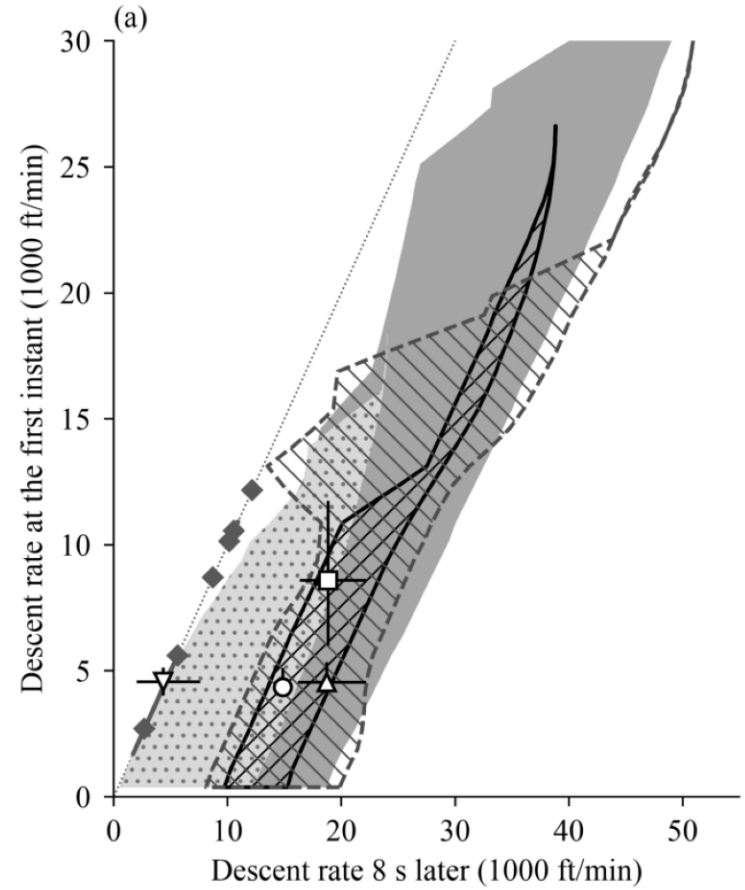
Scorecard: five treatments against evidence, the 18:25 log-on and the dive comparison. Strength: shows in one view that no treatment passes every test. Weakness: three-level symbols at 10 % and 90 % thresholds.



(a) Descent rate at a first instant and 8 s later during dive build-ups; each area covers one group of dives, and the dotted line is a steady descent. Symbols mark the 00:19:29 and 00:19:37 rates implied by each 00:19 BFO treatment; bars are 5–95 % posterior ranges (Davey model with fuel), centred on the published point for Holland transient bias. (b) Tests of each treatment: ln B against the BFOs as observed (only for treatments with both BFOs); fit when applied to the 18:25 log-on (half: fit only at the top of Holland's range, which was set from that log-on); posterior weight inside the areas (filled ≥ 90 %, half 10–90 %, open < 10 %). R600 and R1200 are the 600 and 1200 bit/s channels.

Candidate A4: descent-0019-A4-compact-table

Compact numeric table (ln B, probability, effective samples) in the right-hand space. Strength: exact values. Weakness: no visual encoding; repeats numbers in the text.



(b)

	ln B (90 % interval)	Posterior probability	Effective samples
○	0 (reference)	0.33	508*
□	+0.31 (+0.18, +0.46)	0.45	3,347
△	−0.43 (−0.60, −0.27)	0.22	1,357*
▽	–	–	22,056
BTO only	–	–	499,760

Push-overs: EgyptAir 990, Atlas Air 3591, SilkAir 185

Roll-offs/upsets: TWA 841, Adam Air 574, AirAsia 8501, China Eastern 5735, Lauda Air 004

Boeing 777 simulations, alternate electrical configuration (simulations 3, 4, 6, 10)

Boeing 777 simulations, normal electrical configuration (simulations 1, 2, 5, 7, 8, 9)

Steady descents and stalls (9 accident cases)

○ 00:19 BFOs as observed

□ 00:19 BFOs, Holland transient bias added

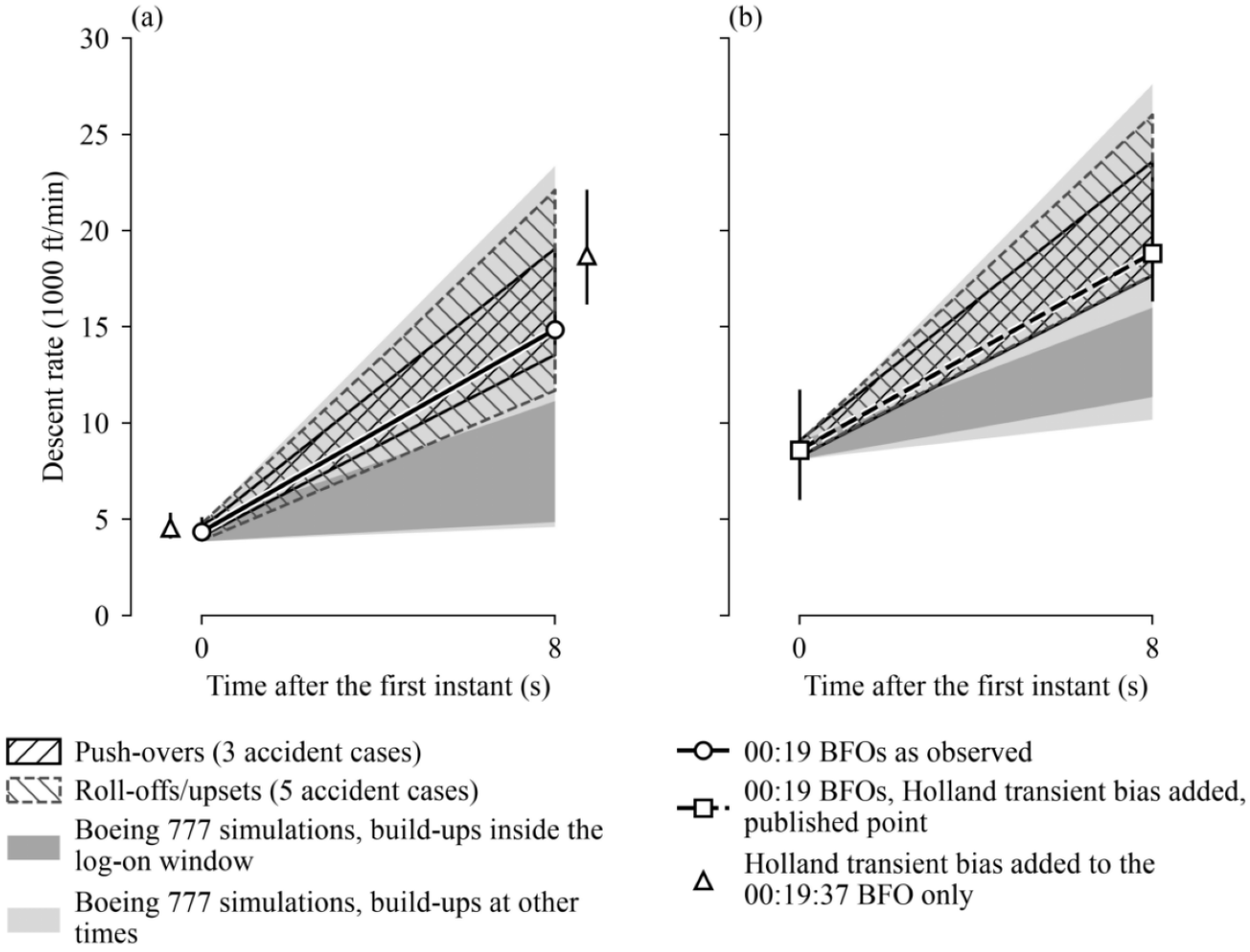
△ Holland transient bias added to the 00:19:37 BFO only

▽ 00:19:29 BFO only, as observed (R600)

(a) Descent rate at a first instant and 8 s later during dive build-ups; each area covers one group of dives, and the dotted line is a steady descent. Symbols mark the 00:19:29 and 00:19:37 rates implied by each 00:19 BFO treatment; bars are 5–95 % posterior ranges (Davey model with fuel), centred on the published point for Holland transient bias. (b) ln B against the BFOs as observed (90 % interval over seeds) and posterior probability with an equal prior, both only for treatments with both BFOs; effective sample size of the final run (* fewer than 500 in the no-pilot family). R600 is the 600 bit/s channel of the 00:19:29 burst.

Candidate B1: descent-0019-B1-slope-bands

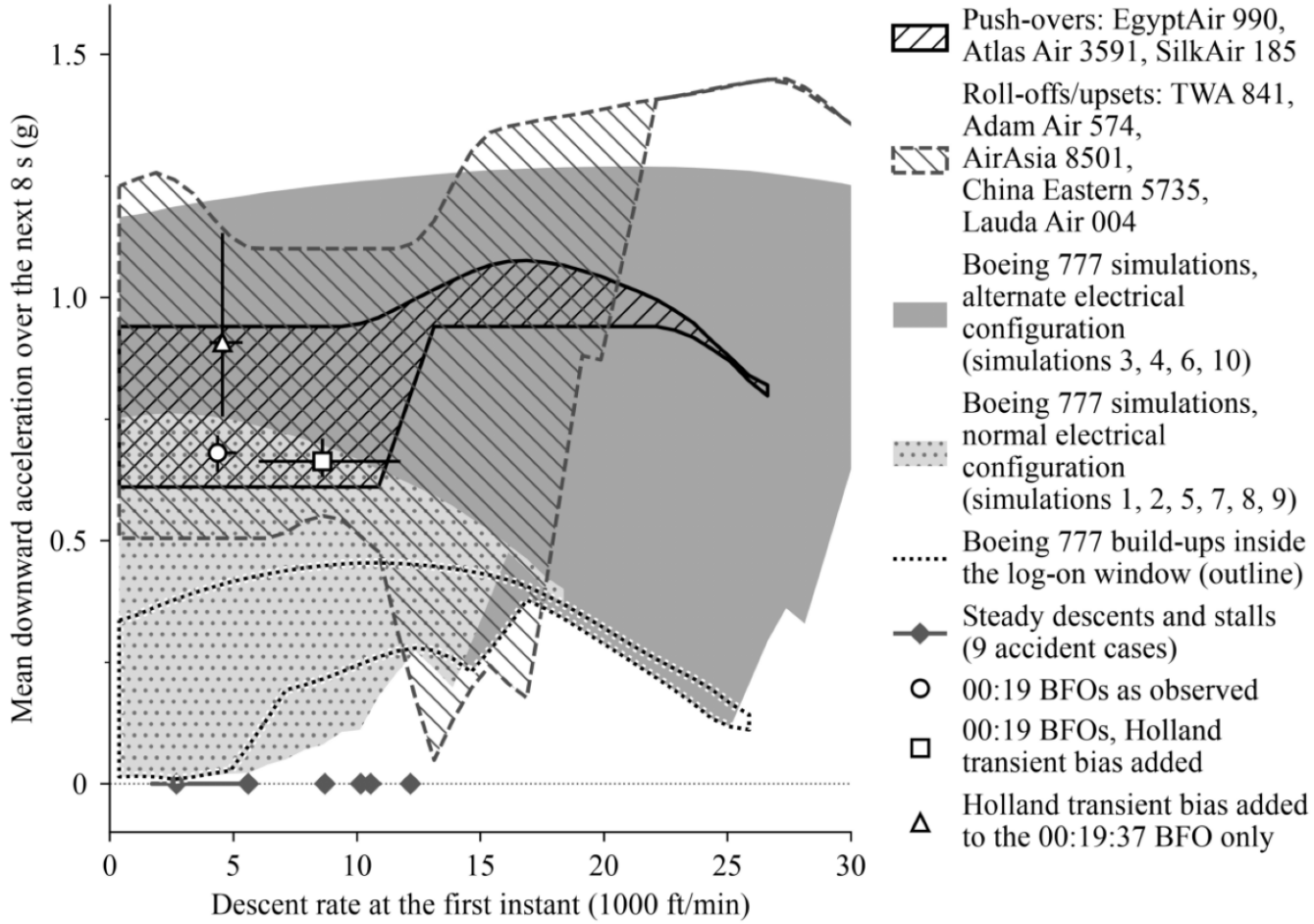
Slope bands: all 8-s build-ups that start within 500 ft/min of each 00:19:29 rate. Strength: direct, uncrowded; the Boeing log-on-window band lies below the implied rise. Weakness: conditional on the 500 ft/min window.



Change of descent rate over 8 s in every dive build-up whose first-instant rate lies within 500 ft/min of the 00:19:29 rate implied by (a) the 00:19 BFOs as observed (4,350 ft/min) and (b) the BFOs with Holland transient bias added (8,590 ft/min, published point). Each area spans all such changes in one group. Heavy lines join the implied rates; bars are 5–95 % posterior ranges at each instant (Davey model with fuel), and triangles give the bias added to the 00:19:37 BFO only. The log-on window is 67–198 s after power loss.

Candidate B2: descent-0019-B2-acceleration-plane

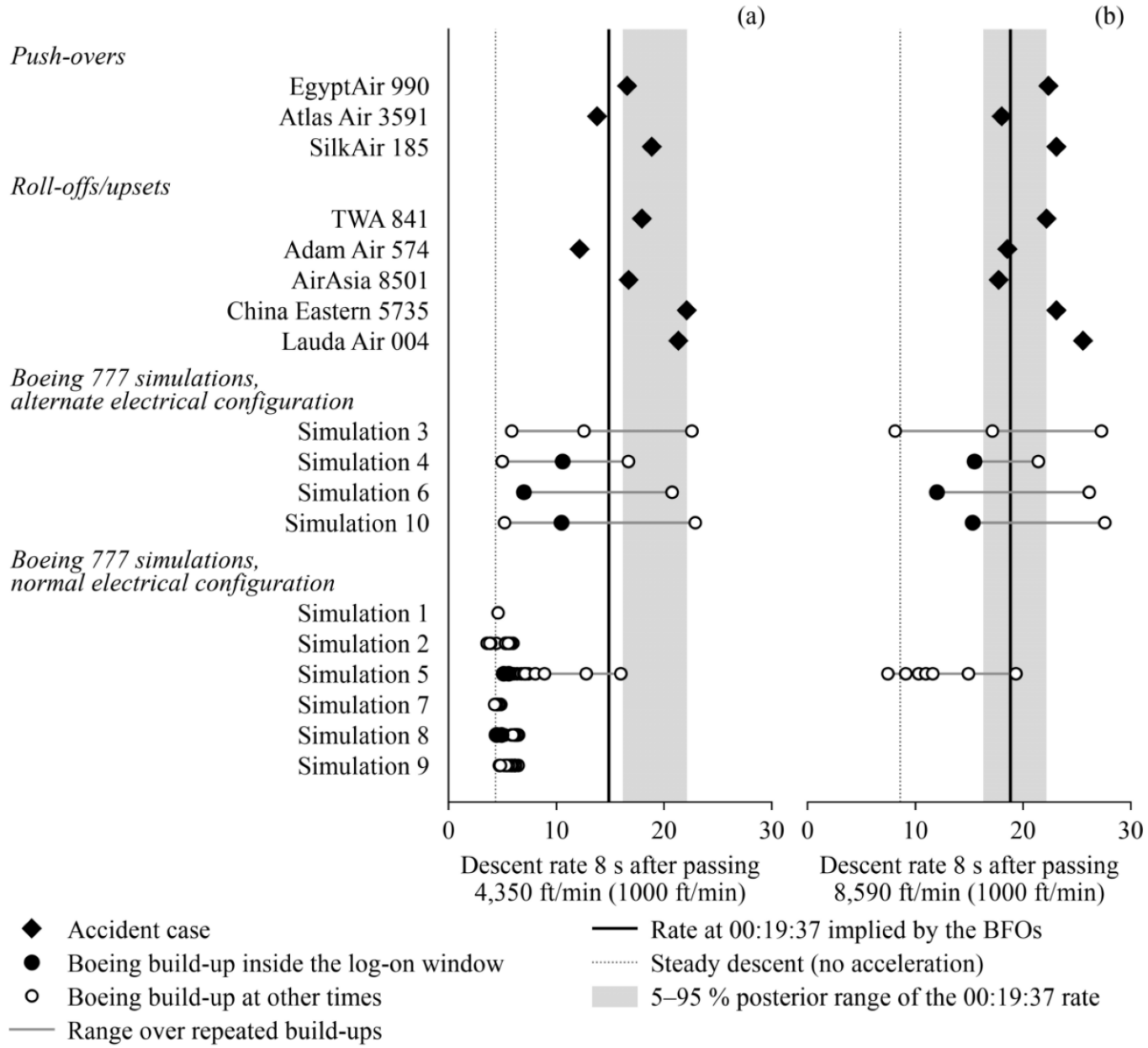
Rate against 8-s mean acceleration (g). Strength: a physical quantity; 0.68 g and 0.66 g lie above the log-on-window outline (at most about 0.45 g). Weakness: constant-acceleration estimates give flat envelope edges.



Descent rate at a first instant and mean downward acceleration over the next 8 s (g = 32.174 ft/s²) during the build-up of dives; each area covers one group, and steady descents lie on the dotted zero line. The dotted outline encloses Boeing build-ups inside the log-on window (67–198 s after power loss). Symbols are the values implied by the 00:19 BFO treatments (0.68 g as observed; 0.66 g with Holland transient bias added, published point); bars give the 5–95 % posterior range (Davey model with fuel).

Candidate B3: descent-0019-B3-conditional-cut

Rate 8 s after each named flight and simulation passed the two 00:19:29 rates. Strength: per-flight, unsmoothed, checkable. Weakness: only two thresholds.



Descent rate 8 s after each dive passed the 00:19:29 rate implied by (a) the 00:19 BFOs as observed (4,350 ft/min) and (b) the BFOs with Holland transient bias added (8,590 ft/min, published point). Solid lines: implied 00:19:37 rates (14,860 and 18,830 ft/min). Grey bands: 5–95 % posterior range of the 00:19:37 rate with (a) the bias added to the 00:19:37 BFO only and (b) the bias added to both (Davey model with fuel). Filled circles are Boeing build-ups inside the log-on window (67–198 s after power loss); a row without a symbol never reaches that rate.